

The ethics of aware and collective artificial intelligence systems

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Abstract

Text.

Introduction

Artificial intelligence (AI) research is advancing rapidly and ethicists—particularly those focused on developing policies to regulate this new technology—are working hard to keep pace (Coeckelbergh, 2020; Nyholm, 2022). In this paper, we argue for the need to further expand the field of AI ethics to cover two new developments in robotics and AI: the shift from singular to collective systems and from non-aware to aware AI. Specifically, we identify several risks and opportunities that should be considered before the general public begins interacting with collective (rather than singular) and/or aware or partially aware (rather than entirely non-aware) AI systems.

Most work in AI ethics to date has focused on human dealings with non-aware systems—or systems that have so far been considered non-aware. This is largely because *awareness* has traditionally been construed as closely tied to *consciousness* (e.g., Block, 1995; Seth and Bayne, 2022). Some notable exceptions aside (e.g., Chalmers, 1995), most researchers agree that present-day technology is far from achieving consciousness (McDermott, 2007; Li et al., 2021; Chalmers, 2023). That said, several large-scale interdisciplinary research projects are currently exploring ways to reconceptualize what it means for a human or a

machine to be aware.¹ Importantly, some of these projects explicitly distinguish machine awareness from machine consciousness and focus specifically on the former (e.g., Bacciu et al., 2025). These developments—and we will argue that they may be fruitful—call for a preemptive and fresh ethical examination of human interaction with AI systems that are introduced to human stakeholders in terms of novel concepts of awareness.

Also, while AI ethics has been predominantly addressing challenges arising in human dealings with individual or singular AI systems, futuristic visions about the impending age of the “internet of things,” along with rapid advances in swarm robotics and distributed AI systems, call for exploration of the ethics of human interaction with collective (rather than singular) AI systems. Add a concise overview of swarm robotics research.

In summary, our aim is to identify ethically relevant questions that arise in human dealings with collective and/or aware AI systems—questions that have not yet been addressed in the field of AI ethics (Figure 1). Our analysis of identified risks and opportunities is by no means exhaustive. The goal is merely to spearhead the broadening of the ethical discourse to encompass developments in robotics and AI.

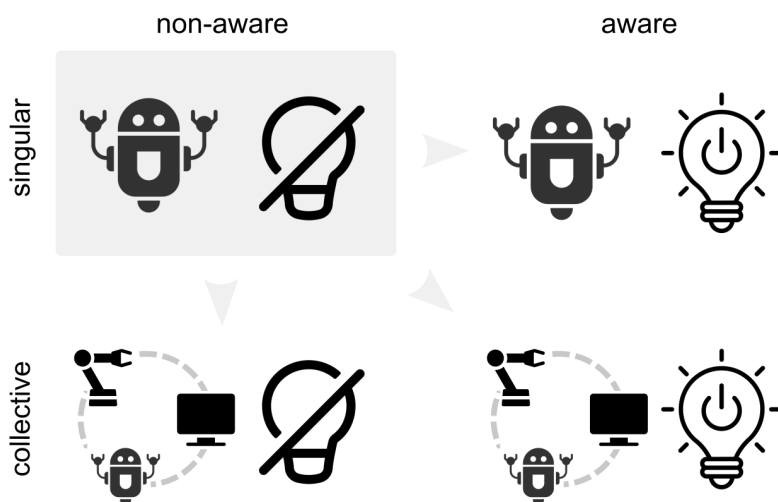


Figure 1. Two avenues of novelty. We consider a shift from singular to collective and from non-aware to aware systems in human dealings with AI. Aware versus non-aware is not necessarily a binary distinction. Awareness may be graded and multidimensional. Also, a shift from singular to collective systems can be further nuanced. For example, a collective system may consist of several connected

¹ A notable example is the suite of research projects funded by the European Innovation Council under the Pathfinder Challenge program titled ‘Awareness Inside’: <https://awarenessinside.eu/>.

individually aware components. Alternatively, collective awareness may be an emergent property of the system as a whole that supervenes on properties of its individual components.

In the examples we discuss to motivate new ethical considerations, we focus on four key topics—cooperation, responsibility, communication, and explainability (Figure 2)—three of which are familiar from existing works on the ethics of human interaction with singular and non-aware systems. They are recognized in current regulatory debates, for example, those surrounding the European Union’s Artificial Intelligence Act (European Commission, 2021). However, human willingness to cooperate with intelligent machines is seldom addressed and thus represents a novel concern. The AI Act, like most regulatory frameworks, primarily focuses on safeguarding human users and those indirectly affected by deployed AI systems. It emphasizes the risks of vulnerable populations being exploited by AI (e.g., O’Neil, 2016), but does not address the reverse form of exploitation: opportunities for humans to exploit the cooperative or safety-oriented design features of interactive AI systems. We believe this is an overlooked and underexplored issue that is important to address for the successful integration of interactive AI systems into society.

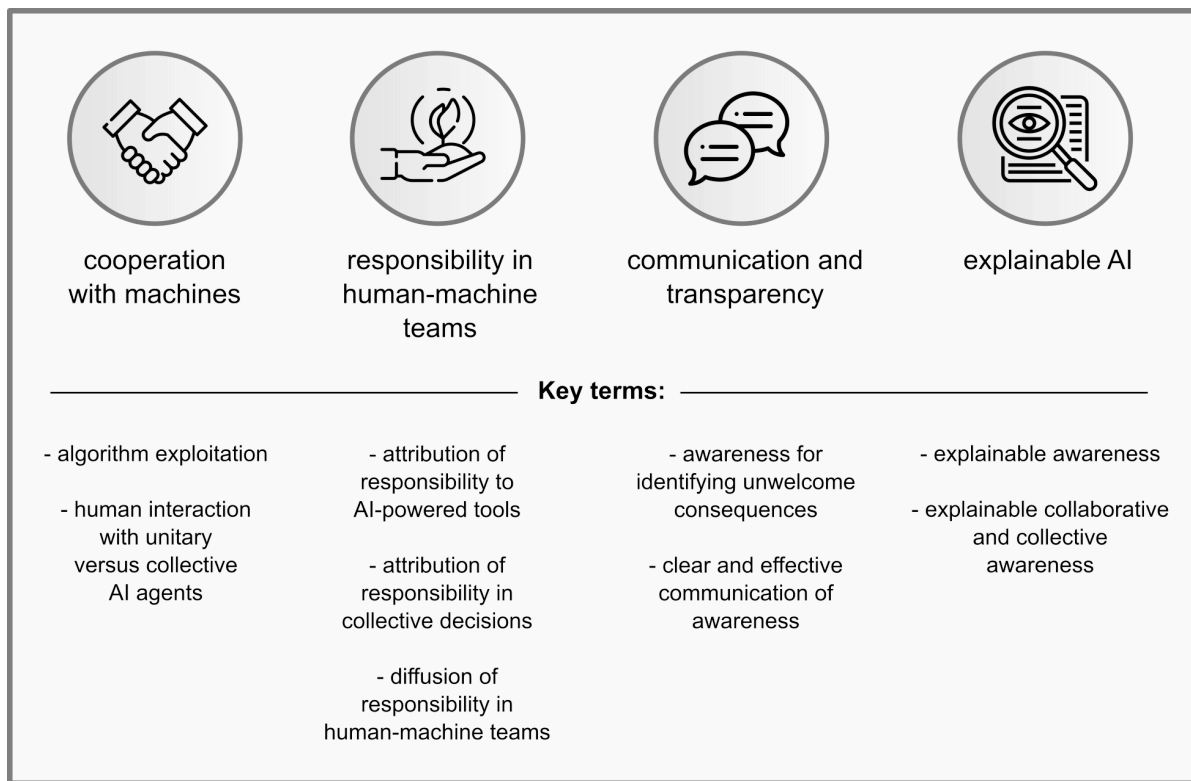


Figure 2. Four areas of focus.

We will address the four topics in turn and provide recommendations for future research on each in the context of aware and/or collective AI systems. First, however, we briefly cover some preliminaries: we clarify our terminology, introduce one proposed approach to disentangling awareness from consciousness, and outline several benefits of adopting this distinction in practice.

Preliminaries

Terminology

We use the term *AI agent* to refer to an AI system that interacts with a human, possibly functioning as an independent decision-making entity. For example, it might be a fully automated tool that assists its human user with a task, or it could be someone else's tool with which the human engages as a co-worker, collaborator, or competitor. We use the terms *unitary*, *singular*, and *individual* AI interchangeably to refer to an AI agent that pursues a goal as a single decision-making unit. In contrast, we use the term *collective* AI to describe an AI agent that operates as part of a larger group of agents jointly pursuing a shared objective. For instance, a fully automated (self-driving) taxi might be designed to safely and efficiently transport as many passengers as possible within its service area in a given timeframe. The optimal strategies it might "learn" to achieve this goal as an individual unit could differ significantly from those it would develop if it were part of a coordinated fleet working collectively toward the same objective.

Awareness as distinct from consciousness

Philosophical debate and everyday usage of the terms often treat *awareness* and *consciousness* interchangeably, or see awareness as an aspect of consciousness (Zeman, 2006; Seth and Byne, 2022). Chalmers (1995) challenged this tendency, proposing awareness to be a somewhat simpler notion than consciousness. Awareness, according to him, allows one to explain what he called "easy problems," for example, an agent's ability to discriminate, categorise, and react to environmental stimuli. The difficulty in disentangling awareness from consciousness, however, remains due to their ambiguous usage to date (Black, 2017). Theories of consciousness vary widely and range from providing fairly minimal to highly complex and demanding concepts of that state (the state of being conscious). The science about consciousness, therefore, addresses different challenges depending on one's chosen theory and definition. Taking consciousness in the phenomenal sense, for example, refers to subjective experiences or 'what it is like' to experience something (Nagel, 1974). In essence, this concerns the mental states or the 'inner life' of a

conscious agent. When it comes to artificial agents, however, it is worthwhile to look at and assess their abilities and capacities whilst remaining neutral on the possibility of them having ‘inner lives,’ adopting the distinction between awareness and consciousness.

The extent to which systems can engage with their environment and other systems varies. The concept of awareness is useful to explore this variation, thereby allowing one to look at, for instance, the detection, registration, and responsiveness of an artificial agent to external environmental events or changes in its internal states. This could reveal the type of strategies a system can employ in the attainment of its goals and the space of action-perception capacities that are available to it. Take, for instance, a vacuuming robot like Roomba. We can ask how aware it is of the space it is navigating through, and the type of objects and other agents that it may encounter in that space, by assessing how successfully it can adapt to changes in its environment (for example, the introduction of a new couch or the presence of another moving agent) without being interested in what, if anything, it might feel like for the robot to be able to do just that. Crucially, by disentangling awareness from consciousness we can sidestep the debate concerning artificial agents’ consciousness—a property that is often reserved for living beings—but nevertheless have a useful and generalizable concept to discuss and assess their capacities to successfully perform their tasks when environments and circumstances change (for more on this idea, see McDermott, 2007; Li et al., 2021; Chalmers, 2023).

Based on these considerations, one research project from those we mentioned earlier is developing a functional account of machine awareness that is best construed as distinct from consciousness (Bacciu et al., 2025). It is a functional account because, since awareness is associated with an agent’s capacities to perform certain tasks, the agent’s awareness itself can be assessed and measured in terms of that agent’s relevant experimentally testable and observable performance in varying environments and circumstances. This is roughly in line with the behaviourist approach in AI ethics research to assess and determine what status could or should be attributed to machines based on their performance in relevant domains, the general idea of which was originally proposed by Alan Turing, but has been developed further since and most recently advocated in AI ethics circles by Danaher (2019, 2021; see also Nyholm, 2022, chapters 8 and 9). This also presents the opportunity to conceptualise awareness as multi-dimensional and graded, which can be of great use in practice. For example, a robot described to have a high degree of spatial awareness of the environment that it navigates could be expected and relied on to perform certain tasks more efficiently and more robustly compared to a robot described to have a low degree of awareness in the relevant (spatial) dimension.

Benefits of machine awareness

There are numerous potential benefits from developing and introducing the concept of machine awareness to people's interactions with AI. One of them we already mentioned. Knowing an AI system's awareness dimensions and its degrees of awareness in those dimensions should make it easier for human users of that system to rely on and delegate tasks to it. For example, knowing that one's fully automated (self-driving) vehicle has a high degree of awareness of the variety of traffic participants it may encounter—pedestrians, cyclists, other vehicles, careless children playing ball close to schools, and so on—would make it easier for one to delegate the task of driving to it in busy urban traffic. Conversely, knowing that one's conversational AI assistant has a low degree of awareness of the truthfulness of its boldly stated claims on a particular topic would make one rightly more sceptical about the assistant's communicated "facts."

Relatedly, accurately and clearly communicated awareness dimensions of a machine, and its degrees of awareness in those dimensions, will allow its human users and supervisors to have a better understanding of where human oversight is the most important and required in collaborative human-machine teams. For example, suppose that one's job requires one to speedily disseminate important information to a wide audience in multiple languages and that one does that with the help of an automated translation tool. Knowing gaps in the translation tool's awareness of important nuances in specific languages will make it easier for one to know translations of which topics and into which languages will require a more careful human oversight and checking (and concerning which topics and languages the tool can be safely relied upon in this collaborative work).

Machine awareness can also safeguard people from unintended and unwelcome side effects of automation. An automated system's awareness of potential problems associated with factors on which it bases its decisions and that therefore contribute to its performance success, whether in the domain of recommending fitness training plans to human users or in filtering job applicants' CVs in a human resources department, will allow it to flag those problematic cases to its human supervisors and thereby help avoid potentially disastrous outcomes. This is particularly relevant in the context of automated nudging of human behaviours (Sætra, 2019; Schmauder et al., 2023). Simply put, the problem with unaware AI systems is that, without being aware of how and why exactly they achieve their tasked objectives, they may produce unintended side effects without anyone noticing. Crucially, in order to be able to flag a potential problem with what one does, one needs to be aware of that potential problem in the first place.

Machine awareness also brings about several important and useful implications for learning. An important dimension of awareness is awareness of one's agency. This enables an agent to assign credit to various factors, for example, in order to distinguish between what was caused by the agent itself and what was caused by other parties or environmental events (other agents, natural phenomena such as wind, and so on). Credit assignment is particularly useful for any agent that needs to "learn" things about its environment and possible courses of action through trial and error.

Another important dimension of awareness is confidence—a form of cognition about cognition, i.e., metacognition. Confidence refers to an agent's sense of uncertainty about the present and future states of the agent's environment and its probabilistic estimates for future outcomes associated with different possible courses of action. Confidence is important for learning because it allows the agent to evaluate its observations and decide whether they are critical enough for the agent to adapt to. An agent endowed with such metacognitive awareness can choose what to adapt to and what can be ignored based on its level of confidence.

In summary, introducing the multi-dimensional and graded term *awareness* to describe machines' behavioural capacities to their users, monitors, and regulators is fruitful in promoting effective and efficient performance of human-machine teams. On the flip side, the anthropomorphic nature of the term *awareness* can give rise to unwanted, ethically relevant consequences, which, among other factors, we address in the following sections.

Cooperation

Text.

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